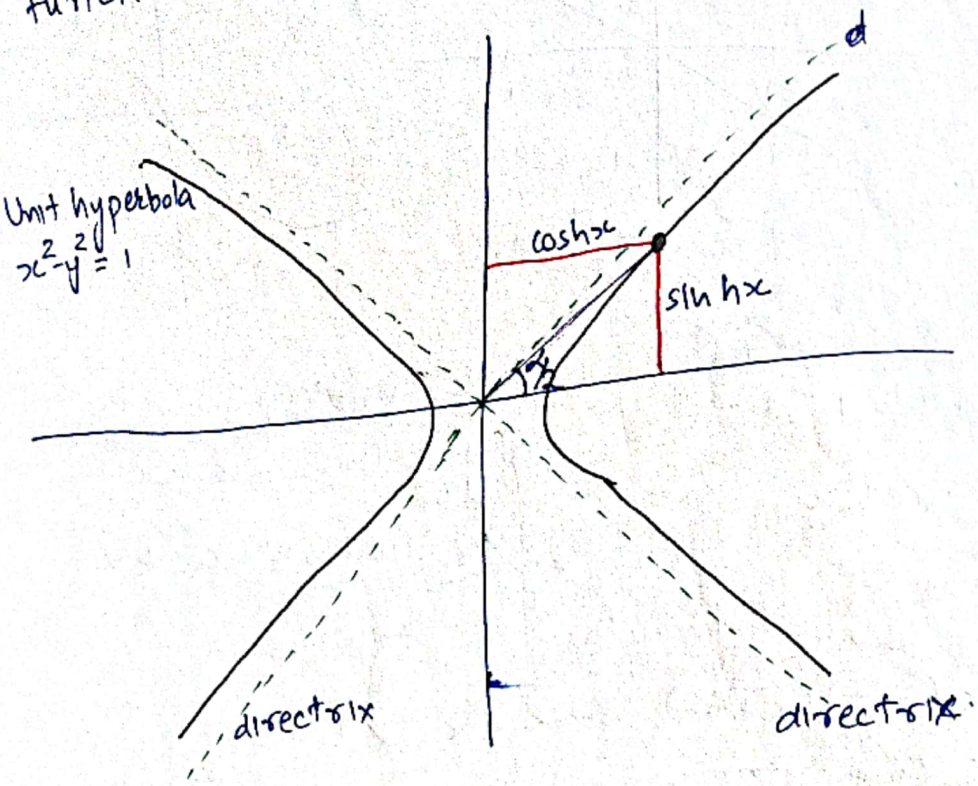
Hyperbolic functions:

A function derived from hyperbola.



Definition:

Function of an angle expressed as a relationship between the distance of a point on a hyperbola to the origin and to the co-ordinate axes.

- Written as sin hx or cos hx.
- Expressed as combination of exponential functions.

Hyperbolic: $\sin hx = \frac{e^x - e^{-x}}{2}$

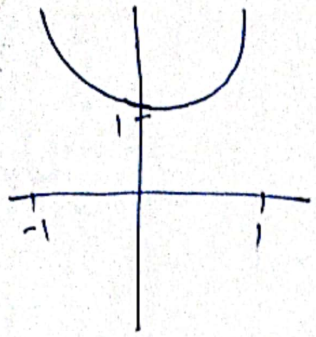
$\cos hx = \frac{e^x + e^{-x}}{2}$

$\tan hx = \frac{\sin hx}{\cos hx} = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^{2x} - 1}{e^{2x} + 1}$

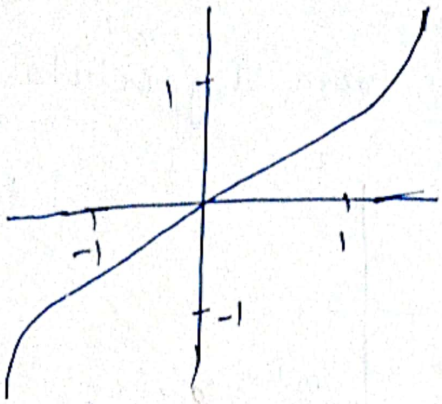
- Range of cos hx is $[1, \infty)$
  sin hx is all real numbers
  tan hx is $(-1, 1)$

Scanned with CamScanner

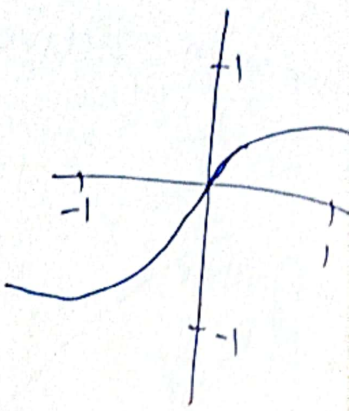
cos h x



sin h x



tan h x



Formulas:

1) $\cosh^2 x - \sinh^2 x = 1$

2) $1 - \tanh^2 x = \operatorname{sech}^2 x$

Derivatives:

1) $\frac{d}{dx} \cosh x = \sinh x$

2) $\frac{d}{dx} \sinh x = \cosh x$

Hyperbolic inverse formulas:

1) $\sinh^{-1} x = \ln\left(x + \sqrt{x^2+1}\right), \quad (-\infty < x < \infty)$

2) $\cosh^{-1} x = \ln\left(x + \sqrt{x^2-1}\right), \quad x \geq 1$

3) $\tanh^{-1} x = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right), \quad -1 < x < 1.$

Inverse formula for trigonometric functions:

1) $\frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}}$

2) $\frac{d}{dx} \cos^{-1} x = -\frac{1}{\sqrt{1-x}}$

3) $\frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2}.$

4) $\sin^{-1} x = x + \frac{1}{2}\frac{x^3}{3} + \frac{1 \cdot 3}{2 \cdot 4}\frac{x^5}{5} + \cdots$

5) $\cos^{-1} x = \frac{\pi}{2} - \sin^{-1} x$

) $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots$

4) $\sinh^{-1} x = x - \frac{x^3}{6} + \frac{3x^5}{40} - \cdots$

)

Scanned with CamScanner

(5) Expansion of $\sin hx$

$$\sin hx = \frac{e^x - e^{-x}}{2} \quad \text{(Refer notes below)}$$

substituting exponential series e^x & e^{-x},

$$\sin hx = \frac{\left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots\right) - \left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots\right)}{2}$$

$$= \frac{2x + 2\frac{x^3}{3!} + 2\frac{x^5}{5!} + \cdots}{2}$$

$$= x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots$$

(6) Expansion of $\cos hx$.

$$\cos hx = \frac{e^x + e^{-x}}{2}$$

substituting e^x & e^{-x},

$$\cos hx = \frac{\left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots\right) + \left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots\right)}{2}$$

$$= \frac{2 + 2\frac{x^2}{2!} + 2\frac{x^4}{4!} + \cdots}{2}$$

$$= 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots$$

Notes:-

1) $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$

2) $e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots$

3) $e^{ax} = 1 + ax + a^2x^2/2! + a^3x^3/3! + \cdots$

Scanned with CamScanner

7) Expansion of $\tanh x$.

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^x - \frac{1}{e^x}}{e^x + \frac{1}{e^x}}$$

$$\therefore \tanh x = \frac{e^{2x} - 1}{e^{2x} + 1}$$

$$= \frac{1 + (2x) + \frac{(2x)^2}{2!} + \frac{(2x)^3}{3!} + \cdots - 1}{1 + (2x) + \frac{(2x)^2}{2!} + \frac{(2x)^3}{3!} + \cdots + 1}$$

$$= \frac{2\left\{x + \frac{2x^2}{2!} + \frac{2^2x^3}{3!} + \cdots\right\}}{2\left[1 + x + \frac{2x^2}{2!} + \frac{2^2x^3}{3!} + \cdots\right]}$$

$$= \frac{x + \frac{x^2}{2} + \frac{2x^3}{3} + \cdots}{1 + x + x^2 + \frac{2x^3}{3} + \cdots}$$

$$= \frac{x\left[1 + x + \frac{2}{3}x^2 + \cdots\right]}{1 + x + x^2 + \frac{2}{3}x^3 + \cdots}$$

Scanned with CamScanner

$$\tanh x = x - \frac{x^3}{3} + \frac{2}{15}x^5 - \cdots$$

$$(1+x)^{-1} = 1 - x + x^2 - x^3 + \cdots$$

$$\tanh x = \frac{\sinh x}{\cosh x}$$

$$= \frac{x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7} + \cdots}{1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \cdots}$$

$$= \left(x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \cdots\right)\left(1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \cdots\right)^{-1}$$

$$\tanh x = \left(x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \cdots\right)\left[1 - \left(\frac{x^2}{2!} + \frac{x^4}{4!} + \cdots\right) + \left(\frac{x^2}{2!} + \frac{x^4}{4!} + \cdots\right)^2 - \left(\frac{x^2}{2!} + \frac{x^4}{4!} + \cdots\right)^3\right]$$

$$= \left(x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots\right)\left[1 - \left(\frac{x^2}{2!} + \frac{x^4}{4!} + \cdots\right) + \left(\frac{x^4}{4} + 2\left(\frac{x^2}{2!}\right)\left(\frac{x^4}{4!}\right) + \cdots\right) - \left\{\frac{x^6}{8} + 3\left(\frac{x^2}{2!}\right)\left(\frac{x^4}{4!}\right) + \left(\frac{x^4}{4!}\right)^3 + \cdots\right.\right.$$

$$= \left(x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots\right) - \left(\frac{x^3}{2!} + \frac{x^5}{4!} + \cdots\right) + \left(\frac{x^5}{4} + \cdots\right)$$

$$= \left(x + \frac{x^3}{6} + \frac{x^5}{120} - \frac{x^3}{2} + \frac{x^5}{24} + \frac{x^5}{4} + \cdots\right)$$

$$= \left\{x - \frac{x^3}{2} + \frac{2}{15}x^5 - \cdots\right.$$

Scanned with CamScanner

Expansion of $\tan x$ by expansion method:

$(1-x)^{-1} = 1+x+x^2+x^3+\cdots$

Let, $y = \tan x = \frac{\sin x}{\cos x}$

$$= \frac{x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots}{1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots}$$

$$= \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right)\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots\right)^{-1}$$

$$= \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right)\left[1 + \left(\frac{x^2}{2!} + \frac{x^4}{4!} + \cdots\right) + \left(\frac{x^2}{2!} + \frac{x^4}{4!} + \cdots\right)^2 + \left(\frac{x^2}{2!} + \frac{x^4}{4!} + \cdots\right)^3 + \right.$$

$$\therefore \tan x = \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right)\left[1 + \left(\frac{x^2}{2!} + \frac{x^4}{4!} + \cdots\right) + \left(\frac{x^4}{4} + 2\frac{x^6}{2!4}\right.\right.$$

$$\left. + \left(\frac{x^6}{2!} + 3\left(\frac{x^2}{2!}\right)\left(\frac{x^4}{4!}\right) + \left(\frac{x^4}{4!}\right)^3 + \cdots\right)\right]$$

$$= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots + \left(\frac{x^3}{2!} + \frac{x^5}{4!} + \cdots\right) + \left(\frac{x^5}{4} + \cdots\right.$$

$$= x - \frac{x^3}{6} + \frac{x^3}{2} + \frac{x^5}{5!} + \frac{x^5}{4!} + \frac{x^5}{4} + \cdots$$

$$= x + \frac{x^3}{3} + \frac{2x^5}{15} + \cdots$$

Scanned with CamScanner

(8) Expansion of $\log(1+x)$

Let, $y = \log(1+x)$

$y(0) = \log 1 = 0$

Now, $y_n = \frac{d^n}{dx^n} \log(1+x) = (-1)^{n-1} \frac{(n-1)!}{(x+1)^n}$

$$\left[y_1 = \frac{1}{1+x} = (-1)^{0} \frac{(0)!}{(1+x)} \right.$$

$$y_2 = (1+x)^{-1} = -1(1+x)^{-2} = -1 \frac{(1)!}{(1+x)^2}$$

$$\left. y_3 = -1(1+x)^{-2} = 2(1+x)^{-3} = (-1)^{3-1} \frac{2!}{(1+x)^3} \right]$$

$$\therefore y_n(0) = (-1)^{n-1}(n-1)!$$

Putting $n = 1, 2, 3, \ldots$

$y_1(0) = 1$, $y_2(0) = -1$, $y_3(0) = 2!$ and so on.

Substituting in Maclaurin's series,

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \ldots$$

This series is known as logarithmic series and is valid for $-1 < x < 1$.

Note:

In above series, replacing x by $-x$,

$$\log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \frac{x^5}{5} - \ldots$$

Scanned with CamScanner

(9) Expansion of $(1+x)^m$ (Binomial series).

Let, $y = (1+x)^m$

$y(0) = (1+0)^m = 1$

$$\begin{cases} y_1 = m(1+x)^{m-1} \\ y_2 = m(m-1)(1+x)^{m-2} \end{cases}$$

$$y_n = m(m-1)(m-2)\cdots(m-n+1)(1+x)^{m-n}$$

$$y_n(0) = m(m-1)(m-2)\cdots(m-n+1)$$

Putting $n = 1, 2, 3, 4, \ldots$

$y_1(0) = m$, $y_2(0) = m(m-1)$, $y_3(0) = m(m-1)(m-2)$ and so on.

Substituting in Maclaurin's series,

$$(1+x)^m = 1 + mx + \frac{m(m-1)}{2!}x^2 + \frac{m(m-1)(m-2)}{3!}x^3 + \cdots$$

This series is known as binomial series and is valid for $-1 < x < 1$.

Scanned with CamScanner

Classroom problems:

By definition:

① Expand 5^x upto the first three non-zero terms of the series.

Let, $f(x) = 5^x$

By Maclaurin's series,

$$f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \cdots \quad — (1)$$

Now, $f(x) = 5^x$, $f(0) = 5^0 = 1$

$f'(x) = 5^x \cdot \log 5$, $f'(0) = \log 5$

$f''(x) = 5^x (\log 5)^2$, $f''(0) = (\log 5)^2$ and so on.

Substituting in eq^n (1),

$$5^x = 1 + x \cdot \log 5 + \frac{x^2}{2!} (\log 5)^2 + \cdots$$

② If $x^3 + y^3 + xy - 1 = 0$, prove that $y = 1 - \frac{x}{3} - \frac{26}{81} x^3 - \cdots$

By Maclaurin's series,

$$y = y(0) + x y_1(0) + \frac{x^2}{2!} y_2(0) + \frac{x^3}{3!} y_3(0) + \cdots \quad (1)$$

$x^3 + y^3 + xy - 1 = 0$ — (2)

Putting $x = 0$, $y^3 - 1 = 0$

$y^3 = 1$

$\therefore y(0) = 1$

Differentiating eq^n (2) w.r. to x,

$3x^2 + 3y^2 y_1 + x y_1 + y = 0$ — (3)

Putting $x = 0$,

$0 + 3y^2(0) y_1(0) + 0 + y(0) = 1$

$3(1) y_1(0) + 1 = 0$

$y_1(0) = -\frac{1}{3}$

Scanned with CamScanner

Differentiating eq^{n} (3) w.r.to x,

$$6x + 6yy_1y_1 + 3y^2y_2 + xy_2 + y_1 + y_1 = 0$$

$$6x + 6yy_1^2 + 3y^2y_2 + xy_2 + 2y_1 = 0 \quad \text{——} (4)$$

Putting $x=0$,

$$0 + 6y(0)y_1^2(0) + 3y^2(0)y_2(0) + 0 + 2y_1(0) = 0$$

$$6(1)\left(-\frac{1}{3}\right)^2 + 3(1)y_2(0) + 2\left(-\frac{1}{3}\right) = 0$$

$$\frac{2}{3} + 3y_2(0) - \frac{2}{3} = 0$$

$$\therefore y_2(0) = 0$$

Differentiating eq^{n} (4) w.r.to x,

$$6 + 6y_1y_1^2 + 6y(2y_1y_2) + 6yyy_1y_2 + 3y^2y_3 + y_2 + xy_3 + y_2$$

$$\therefore 6 + 6y_1^3 + 12yy_1y_2 + 6yyy_1y_2 + 3y^2y_3 + 3y_2 + xy_3 = 0$$

Putting $x=0$,

$$6 + 6y_1^3(0) + 12y(0)y_1(0)y_2(0) + 6y(0)y_1(0)y_2(0) + 3y^2(0)y_3(0) + 3y_2(0) + 0 = 0$$

$$\therefore 6 + \left(-\frac{1}{3}\right)^3 \cdot 6 + 0 + 0 + 3(1)^2 y_3(0) + 0 = 0$$

$$\therefore 6 + 6\left(\frac{-1}{27}\right) + 3y_3(0) = 0$$

$$\therefore 6 - \frac{2}{9} + 3y_3(0) = 0$$

$$\therefore y_3(0) = -\frac{52}{27} \text{ and so on.}$$

Substituting in eq^{n},

$$y = 1 - \frac{x}{3} + \frac{x^2}{2!}(0) + \frac{x^3}{3!}\left(\frac{-52}{27}\right) + \cdots$$

$$= 1 - \frac{x}{3} - \frac{26}{81}x^3 - \cdots$$

Scanned with CamScanner

Obtain the series $\log(1+x)$ and find the series $\log\left(\frac{1+x}{1-x}\right)$ and hence find the value of $\log_e\left(\frac{11}{9}\right)$.

Let, $y = \log(1+x)$

By Maclaurin's series,

$$y = y(0) + \frac{x}{1!}y_1(0) + \frac{x^2}{2!}y_2(0) + \frac{x^3}{3!}y_3(0) + \cdots \quad -(1)$$

$y(0) = \log 1 = 0$

$y_1 = \frac{1}{1+x}, \quad y_1(0) = 1$

$y_2 = \frac{-1}{(1+x)^2}, \quad y_2(0) = -1$

$y_3 = \frac{(2!)}{(1+x)^3}, \quad y_3(0) = 2!$

$y_4 = -\frac{(3!)}{(1+x)^4}, \quad y_4(0) = -(3!)$

Substituting in eq^n (1),

$$y = 0 + x - \frac{x^2}{2!} + \frac{x^3}{3!}(2!) - \frac{x^4}{4!}(3!) + \cdots$$

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$

Replacing x by $-x$,

$$\log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \cdots$$

Now, $\log\left(\frac{1+x}{1-x}\right) = \log(1+x) - \log(1-x)$

$$= \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots\right) - \left[-x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \cdots\right]$$

$$= \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots\right) + \left(x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \cdots\right)$$

$$= 2\left(x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots\right)$$

Putting $x = \frac{1}{10}$ and considering first three terms,

$$\log_e\left(\frac{11}{9}\right) = 2\left[\frac{1}{10} + \frac{1}{3}\cdot\frac{1}{(10)^3} + \frac{1}{5}\left(\frac{1}{10^5}\right)\right] = 0.20067$$

Scanned with CamScanner

By Standard Expansion!

① Prove that $e^{e^x} = e\left(1 + x + x^2 + \frac{5x^3}{6} + \cdots\right)$

$$e^{e^x} = e^{\left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots\right)}$$

$$= e^1 \cdot e^{\left(x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots\right)}$$

$$= e \cdot \left[1 + \left(x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots\right) + \frac{1}{2!}\left(x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots\right)^2 + \frac{1}{3!}\left(x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots\right)^3 + \cdots\right]$$

$$= e\left[1 + x + x^2\left(\frac{1}{2} + \frac{1}{2}\right) + x^3\left(\frac{1}{6} + \frac{1}{2} + \frac{1}{6}\right) + \cdots\right]$$

$$= e\left(1 + x + x^2 + \frac{5}{6}x^3 + \cdots\right)$$

② Prove that $e^{x\sin x} = 1 + x^2 + \frac{x^4}{3} + \frac{x^6}{120} + \cdots$

$$e^{x\sin x} = \left[1 + (x\sin x) + \frac{(x\sin x)^2}{2!} + \frac{(x\sin x)^3}{3!} + \cdots\right]$$

$$= 1 + x\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right) + \frac{x^2}{2!}\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right)^2 + \frac{x^3}{3!}\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots\right)^3$$

$$= 1 + x^2 + x^4\left(-\frac{1}{6} + \frac{1}{2}\right) + x^6\left(\frac{1}{120} - \frac{1}{6} + \frac{1}{6}\right) + \cdots$$

$$= 1 + x^2 + \frac{x^4}{3} + \frac{x^6}{120} + \cdots$$

③ Prove that $\cos x \cdot \cosh x = 1 - \frac{2^2x^4}{4!} + \frac{2^2x^8}{8!} - \cdots$

$$\cos x \cdot \cosh x = \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \cdots\right)\left(1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \frac{x^8}{8!} + \cdots\right)$$

$$= 1 + x^4\left(\frac{1}{4!} - \frac{1}{(2!)^2}\right)$$

Scanned with CamScanner

$$= \left(1+\frac{x^2}{2!}+\frac{x^4}{4!}+\frac{x^6}{6!}+\frac{x^8}{8!}+\cdots\right) - \left(\frac{x^2}{2!}+\frac{x^2}{2!}\frac{x^2}{2!}+\frac{x^2}{2!}\frac{x^4}{4!}+\frac{x^2}{2!}\frac{x^6}{6!}+\cdots\right)$$

$$+\left(\frac{x^4}{4!}+\frac{x^4}{4!}\frac{x^2}{2!}+\frac{x^4}{4!}\frac{x^4}{4!}+\cdots\right) - \left(\frac{x^6}{6!}+\frac{x^6}{6!}\frac{x^2}{2!}+\cdots\right)+\left(\frac{x^8}{8!}+\cdots\right.$$

$$= 1 + x^4\left(\frac{2}{4!}-\frac{1}{2!2!}\right) + x^8\left(\frac{2}{8!}-\frac{2}{2!6!}+\frac{1}{4!4!}\right)+\cdots$$

$$= 1 - \frac{2^2x^4}{4!} + \frac{2^2x^8}{8!} - \cdots$$

$\left[x^4\left(\frac{2}{4!}-\frac{1}{2!2!}\right)\right.$ - In proof we want with power of x^4, $4!$

$\left.\therefore x^4\left(\frac{2}{4!}-\frac{4!}{4!2!2!}\right)\right] = x^4\left(\frac{2}{4!}-\frac{4\cdot3\cdot2}{2\cdot2\cdot4!}\right) = x^4\left(\frac{2-6}{4!}\right)$

$= x^4\left(\frac{-4}{4!}\right) = -\frac{x^4}{4!}2^2$. Ily for x^8, $\frac{2^2x^8}{8!}$ --.

1) Expand $\log(1+x+x^2+x^3)$ upto x^8.

$$\log(1+x+x^2+x^3) = \log\left[(1+x)+x^2(1+x)\right]$$

$$= \log\left[(1+x)(1+x^2)\right] \quad \because \log xy = \log x + \log y$$

$$= \log(1+x) + \log(1+x^2)$$

$$= \left[x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\frac{x^5}{5}-\frac{x^6}{6}+\frac{x^7}{7}-\frac{x^8}{8}+\cdots\right] +$$

$$\left[x^2-\frac{(x^2)^2}{2}+\frac{(x^2)^3}{3}-\frac{(x^2)^4}{4}+\cdots\right]$$

$$= x + x^2\left(1-\tfrac{1}{2}\right)+\frac{x^3}{3}-x^4\left(\tfrac{1}{4}+\tfrac{1}{2}\right)+\frac{x^5}{5}-x^6\left(\tfrac{1}{6}+\tfrac{1}{3}\right)+$$

$$\frac{x^7}{7}-x^8\left(\tfrac{1}{8}+\tfrac{1}{4}\right)+\cdots$$

$$= x+\frac{x^2}{2}+\frac{x^3}{3}-\frac{3}{4}x^4+\frac{x^5}{5}+\frac{x^6}{6}+\frac{x^7}{7}-\frac{3}{8}x^8+\cdots$$

Scanned with CamScanner

⑤ Prove that $\log\left(\frac{\sin hx}{x}\right) = \frac{x^2}{6} - \frac{x^4}{180} + \cdots$

$$\log\left(\frac{\sin hx}{x}\right) = \log\left[\frac{1}{x}\left(x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots\right)\right]$$

$$= \log\left[1 + \frac{x^2}{3!} + \frac{x^4}{4!} + \cdots\right]$$

$$= \left(\frac{x^2}{3!} + \frac{x^4}{5!} + \cdots\right) - \frac{1}{2}\left(\frac{x^2}{3!} + \frac{x^4}{5!} + \cdots\right)^2 + \cdots$$

$$= \left(\frac{x^2}{3!} + \frac{x^4}{5!} + \cdots\right) - \frac{1}{2}\left(\frac{x^4}{(3!)^2} + \cdots\right)$$

$$= \frac{x^2}{6} - x^4\left(\frac{1}{2\cdot 6\cdot 6} - \frac{1}{120}\right) + \cdots$$

$$= \frac{x^2}{6} - x^4\left(\frac{1}{72} - \frac{1}{120}\right) + \cdots$$

$$= \frac{x^2}{6} - x^4\left(\frac{+2}{360}\right) + \cdots$$

$$= \frac{x^2}{6} - \frac{x^4}{180} + \cdots$$

⑥. Prove that $\log x = \log 2 + \left(\frac{x}{2} - 1\right) - \frac{1}{2}\left(\frac{x}{2} - 1\right)^2 + \frac{1}{3}\left(\frac{x}{2} - 1\right)$

$$\log x = \log\left(2 \cdot \frac{x}{2}\right)$$

$$= \log 2 + \log\frac{x}{2}$$

$$= \log 2 + \log\left[1 + \left(\frac{x}{2} - 1\right)\right]$$

$$= \log 2 + \left(\frac{x}{2} - 1\right) - \frac{1}{2}\left(\frac{x}{2} - 1\right)^2 + \frac{1}{3}\left(\frac{x}{2} - 1\right)^3 + \cdots$$

) Prove that $\sin(e^x - 1) = x + \frac{x^2}{2} - \frac{5}{24}x^4 + \cdots$

$$\sin(e^x - 1) = \sin\left[\left\{1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots\right\} - 1\right]$$

$$= \sin\left[x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots\right]$$

$$= \left(x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots\right) - \frac{1}{3!}\left(x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots\right)^3 + \cdots$$

$$= \left(x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots\right) - \frac{1}{3!}\left[x^3 + 3x \cdot \frac{x^2}{2!}\left(x + \frac{x^2}{2!}\right)\right.$$

Scanned with CamScanner

$$= \left(x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots\right) - \frac{1}{3!}\left(x^3 + \frac{3x^4}{2!} + \cdots\right)$$

$$= x + \frac{x^2}{2} + x^4\left[\frac{1}{24} + \frac{3}{4\cdot3\cdot2}\right] + \cdots$$

$$= x + \frac{x^2}{2} + x^4\left[\frac{1}{24} - \frac{1}{4}\right] + \cdots$$

$$= x + \frac{x^2}{2} - \frac{5}{24}x^4 + \cdots$$

Prove that $\tan^{-1}\left(\frac{x\sin\alpha}{1-x\cos\alpha}\right) = x\sin\alpha + \frac{x^2}{2}\sin 2\alpha + \frac{x^3}{3}\sin 3\alpha + \cdots$

Let, $y = \tan^{-1}\left(\frac{x\sin\alpha}{1-x\cos\alpha}\right)$

$$\tan y = \frac{x\sin\alpha}{1-x\cos\alpha}$$

$$\left[\sin y = \frac{e^{iy} - e^{-iy}}{2i},\ \cos y = \frac{e^{iy}+e^{-iy}}{2}\right.$$

$$\therefore \frac{e^{iy} - e^{-iy}}{i(e^{iy} + e^{-iy})} = \frac{x\sin\alpha}{1-x\cos\alpha}$$

$$\therefore \frac{e^{iy} - e^{-iy}}{e^{iy} + e^{-iy}} = \frac{ix\sin\alpha}{1-x\cos\alpha}$$

Applying componendo & dividendo,

$$\frac{e^{iy} - e^{-iy} + e^{iy} + e^{-iy}}{e^{iy} - e^{-iy} - (e^{iy} + e^{-iy})} = \frac{ix\sin\alpha + 1 - x\cos\alpha}{ix\sin\alpha - (1 - x\cos\alpha)}$$

$$\frac{2e^{iy}}{-2e^{-iy}} = \frac{1 - x(\cos\alpha + i\sin\alpha)}{-1 + x\cos\alpha + ix\sin\alpha - 1 + x\cos\alpha}$$

$$\frac{e^{iy}}{-e^{-iy}} = \frac{1 - x(\cos\alpha + i\sin\alpha)}{-[1 - x(\cos\alpha + i\sin\alpha)]}$$

Scanned with CamScanner

(9)

$$\frac{2e^{iy}}{-2e^{-iy}} = \frac{1-x(\cos\alpha \pm i\sin\alpha)}{ix\sin\alpha - 1 + x\cos\alpha}$$

$$\therefore \frac{e^{iy}}{-e^{-iy}} = \frac{1-x(\cos\alpha \pm i\sin\alpha)}{-[1 - ix\sin\alpha - x\cos\alpha]}$$

$$\therefore \frac{e^{iy}}{-e^{-iy}} = \frac{1-x(\cos\alpha \pm i\sin\alpha)}{-[1-x(\cos\alpha + i\sin\alpha)]}$$

$$\therefore \frac{e^{iy}}{e^{-iy}} = \frac{1-x\cdot(\cos\alpha \pm i\sin\alpha)}{1-x(\cos\alpha + i\sin\alpha)}$$

$$\therefore \frac{e^{iy}}{e^{-iy}} = \frac{1-xe^{-i\alpha}}{1-xe^{i\alpha}}$$

$$\therefore e^{2iy} = \frac{1-xe^{-i\alpha}}{1-xe^{i\alpha}}$$

$$\therefore 2iy = \log\left[\frac{1-xe^{-i\alpha}}{1-xe^{i\alpha}}\right]$$

$$= \log(1-xe^{-i\alpha}) - \log(1-xe^{i\alpha})$$

$$= \left(-xe^{-i\alpha} - \frac{x^2}{2}e^{-2i\alpha} - \frac{x^3}{3}e^{-3i\alpha} - \cdots\right) -$$

$$\left(-xe^{i\alpha} - \frac{x^2e^{2i\alpha}}{2} - \frac{x^3e^{3i\alpha}}{3} - \cdots\right)$$

$$= \left(-xe^{-i\alpha} - \frac{x^2}{2}e^{-2i\alpha} - \frac{x^3}{3}e^{-3i\alpha} - \cdots\right) +$$

$$xe^{i\alpha} + \frac{x^2}{2}e^{2i\alpha} + \frac{x^3}{3}e^{3i\alpha} + \cdots$$

$$= x(e^{i\alpha} - e^{-i\alpha}) + \frac{x^2}{2}(e^{2i\alpha} - e^{-2i\alpha}) + \frac{x^3}{3}(e^{3i\alpha} - e^{-3i\alpha}$$

$$\therefore 2iy = x(2i\sin\alpha) + \frac{x^2}{2}(2i\sin 2\alpha) + \frac{x^3}{3}(2i\sin 3\alpha) + \cdots$$

$$\therefore y = x\sin\alpha + \frac{x^2}{2}\sin 2\alpha + \frac{x^3}{3}\sin 3\alpha + \cdots$$

Scanned with CamScanner